
NUMERICAL REPRODUCTION OF GRADIENT FLOW SAMPLERS: FROM GAUSSIAN TO LOG-CONCAVE TARGETS

SI114H GROUP PROJECT REPORT

Ping Xindi

2023533153

pingxd2023@shanghaitech.edu.cn

Hu Xinying

2023533017

huxy2023@shanghaitech.edu.cn

Ma Jingyi

2023533013

majy2023@shanghaitech.edu.cn

June 14, 2026

ABSTRACT

Gradient-flow formulations provide a unified geometric perspective on sampling algorithms for Bayesian inference. Following the framework of Chen et al. [2023], we study sampling dynamics induced by different probability-space geometries and reproduce the main numerical experiments on Gaussian and log-concave target distributions. We implement three Gaussian approximation flows and two non-parametric Wasserstein particle methods, with particular focus on the role of affine invariance in ill-conditioned sampling problems. Our results confirm that affine-invariant dynamics substantially accelerate convergence on anisotropic targets compared with non-invariant Euclidean and Wasserstein baselines. In addition, we propose Normalized Local Affine-Wasserstein (NLAW), a new affine-invariant particle method that augments the global covariance preconditioner with a normalized local rank-one correction. Experiments on the same benchmark problems show that NLAW preserves the robustness of affine-invariant Wasserstein dynamics on Gaussian targets while improving early-stage exploration on non-Gaussian log-concave targets.

Keywords Gradient Flow · Affine-invariance · Sampling from unnormalized densities

1 Introduction

Sampling from posterior distributions is a central problem in Bayesian inference and computational statistics. Standard Langevin dynamics provides a principled MCMC approach, but its efficiency can degrade severely on anisotropic or ill-conditioned targets, where different directions require very different exploration scales.

Chen et al. [2023] address this issue by formulating sampling as a gradient flow over probability measures. Their framework connects Wasserstein, Fisher–Rao, and Stein geometries with both mean-field particle methods and Gaussian approximation flows, and shows that affine-invariant preconditioning can significantly improve convergence on anisotropic distributions.

In this paper, we reproduce the first two benchmark problems from Chen et al. [2023]: a two-dimensional Gaussian posterior and a two-dimensional log-concave posterior. We implement three Gaussian approximation methods and two Wasserstein particle methods, and evaluate them using mean error, covariance error, and projection-function error.

Beyond reproduction, we propose Normalized Local Affine-Wasserstein (NLAW), a local adaptive extension of affine-invariant Wasserstein sampling. NLAW augments the global covariance preconditioner with a normalized rank-one correction based on each particle’s displacement from the ensemble mean. This preserves affine invariance while introducing local adaptivity. Our experiments show that NLAW remains stable on Gaussian targets and improves early-stage exploration on the non-Gaussian log-concave target.

2 Literature Review and Overview of Existing Work

2.1 Markov Chain Monte Carlo and Langevin Dynamics

Bayesian inference aims to sample from a posterior distribution of the form

$$\pi(\theta) \propto \exp(-\Phi(\theta)), \quad \theta \in \mathbb{R}^d,$$

where $\Phi(\theta)$ is a smooth potential function.

A fundamental class of methods for this task is Markov Chain Monte Carlo (MCMC), with Langevin dynamics serving as a canonical continuous-time model:

$$d\theta_t = -\nabla\Phi(\theta_t) dt + \sqrt{2} dW_t.$$

This stochastic differential equation provides an asymptotically exact sampling scheme whose invariant distribution is π Pavliotis [2014]. However, its practical efficiency degrades significantly in the presence of anisotropic target geometries.

Let $H(\theta) = \nabla^2\Phi(\theta)$ denote the Hessian of the potential. When the condition number

$$\kappa = \frac{\lambda_{\max}(H)}{\lambda_{\min}(H)}$$

is large, isotropic noise leads to slow exploration along low-curvature directions. Moreover, stable discretization requires step sizes scaling with $\mathcal{O}(\lambda_{\max}^{-1})$, which severely limits convergence speed in ill-conditioned regimes.

These limitations motivate geometric reformulations of sampling algorithms.

2.2 Sampling as Optimization in Probability Space

An alternative perspective interprets sampling as an optimization problem over the space of probability measures $\mathcal{P}(\mathbb{R}^d)$. In this framework, the target distribution is characterized as the minimizer of an energy functional.

A standard choice is the Kullback–Leibler divergence:

$$\mathcal{E}(\rho) = \text{KL}(\rho\|\pi),$$

which provides a natural measure of discrepancy between distributions Wainwright and Jordan [2008].

This perspective enables the interpretation of sampling as a gradient flow in an infinite-dimensional space of probability measures, where the geometry is determined by the choice of metric structure Santambrogio [2017].

2.3 Geometric Structures for Sampling Dynamics

Different choices of metric on $\mathcal{P}(\mathbb{R}^d)$ induce different sampling dynamics:

- **Wasserstein Geometry (W_2):** Induced by optimal transport, this geometry leads to dynamics equivalent to the Fokker–Planck equation and connects directly to Langevin sampling via gradient flows Jordan et al. [1998].
- **Fisher–Rao Geometry:** Based on information geometry, this structure corresponds to natural gradient flows and captures density evolution without spatial transport.
- **Stein Geometry:** Constructed via reproducing kernel Hilbert spaces (RKHS), it underlies Stein variational gradient descent (SVGD), enabling deterministic particle-based inference.

These viewpoints provide a unified geometric interpretation of sampling algorithms as different gradient flows on probability spaces Santambrogio [2017].

2.4 Limitations and Motivation for Preconditioning

Although Wasserstein-based Langevin dynamics provides a principled geometric interpretation of sampling, it inherits the Euclidean structure of the ambient space. As a result, it is sensitive to affine transformations and poorly conditioned target distributions.

In particular, anisotropic potentials induce highly skewed Hessian spectra, which significantly slow down exploration in low-curvature directions Pavliotis [2014].

To address this issue, recent approaches introduce geometry-adaptive preconditioning strategies that modify the underlying metric according to the evolving distribution. A particularly effective approach is covariance-based preconditioning, which yields affine-invariant sampling dynamics Goodman and Weare [2010], Carrillo and Vaes [2021].

These ideas form the foundation for the affine-invariant gradient flow and particle system developed in Section 3.

3 Integration of the Course Content with the Questions Studied

Building upon the gradient-flow formulation of sampling introduced in Section 2, we derive geometry-aware sampling dynamics in continuous time and their corresponding particle discretizations. The development follows the Wasserstein gradient flow framework Jordan et al. [1998], Santambrogio [2017] and its stochastic counterpart via Langevin dynamics Pavliotis [2014]. We further extend this formulation to covariance-preconditioned dynamics that exhibit affine invariance ??.

3.1 Wasserstein Gradient Flow and Langevin Dynamics

Let $\rho_t \in \mathcal{P}_2(\mathbb{R}^d)$ evolve under the L^2 -Wasserstein gradient flow:

$$\partial_t \rho_t = \nabla \cdot (\rho_t v_t), \quad v_t = -\nabla \frac{\delta \mathcal{E}(\rho_t)}{\delta \rho_t}, \quad (1)$$

where $\mathcal{E}(\rho) = \text{KL}(\rho \parallel \pi)$ is the energy functional.

The first variation is given by Jordan et al. [1998]:

$$\frac{\delta \mathcal{E}}{\delta \rho} = \log \rho + \Phi + 1. \quad (2)$$

Substituting into the velocity field yields:

$$v_t = -\nabla \log \rho_t - \nabla \Phi. \quad (3)$$

This leads to the Fokker–Planck equation Pavliotis [2014]:

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla \Phi) + \Delta \rho_t. \quad (4)$$

By the Itô–Fokker–Planck correspondence Pavliotis [2014], the equivalent particle dynamics is given by the overdamped Langevin SDE:

$$d\theta_t = -\nabla \Phi(\theta_t) dt + \sqrt{2} dW_t. \quad (5)$$

3.2 Affine Transformation and Ill-Conditioning

Consider an affine transformation:

$$\tilde{\theta} = A\theta + b, \quad A \in \text{GL}(d, \mathbb{R}). \quad (6)$$

The push-forward density satisfies:

$$\tilde{\Phi}(\tilde{\theta}) = \Phi(A^{-1}(\tilde{\theta} - b)) + \text{const}. \quad (7)$$

Gradients transform as:

$$\nabla_{\theta} \Phi(\theta) = A^T \nabla_{\tilde{\theta}} \tilde{\Phi}(\tilde{\theta}). \quad (8)$$

When the potential is highly anisotropic, the Hessian spectrum becomes ill-conditioned, resulting in slow mixing for isotropic Langevin dynamics Pavliotis [2014].

3.3 Covariance Preconditioning and Adaptive Geometry

To mitigate anisotropy, we introduce a state-dependent metric tensor defined by the covariance of the evolving distribution:

$$C(\rho_t) = \mathbb{E}_{\rho_t}[(\theta - m_t)(\theta - m_t)^T], \quad m_t = \mathbb{E}_{\rho_t}[\theta]. \quad (9)$$

We define the covariance-preconditioned Langevin dynamics:

$$d\theta_t = -\nabla \Phi(\theta_t) C(\rho_t) dt + \sqrt{2C(\rho_t)} dW_t. \quad (10)$$

Remark (Implementation convention). In the numerical implementation (Section 5), gradients are represented as row vectors, and covariance acts via right multiplication, ensuring consistency with the Euler–Maruyama particle update.

3.4 Affine Invariance of the Preconditioned Dynamics

Under $\tilde{\theta} = A\theta + b$, the covariance transforms as Goodman and Weare [2010], Carrillo and Vaes [2021]:

$$C(\tilde{\rho}_t) = AC(\rho_t)A^\top. \quad (11)$$

Applying the transformation yields:

$$d\tilde{\theta}_t = Ad\theta_t = -A\nabla\Phi(\theta_t)C(\rho_t)dt + \sqrt{2}A\sqrt{C(\rho_t)}dW_t. \quad (12)$$

For diffusion:

$$(A\sqrt{C})(A\sqrt{C})^\top = ACA^\top = C(\tilde{\rho}_t), \quad (13)$$

implying:

$$\sqrt{2}A\sqrt{C(\rho_t)}dW_t = \sqrt{2C(\tilde{\rho}_t)}d\tilde{W}_t. \quad (14)$$

Thus:

$$d\tilde{\theta}_t = -C(\tilde{\rho}_t)\nabla_{\tilde{\theta}}\tilde{\Phi}(\tilde{\theta}_t)dt + \sqrt{2C(\tilde{\rho}_t)}d\tilde{W}_t, \quad (15)$$

which establishes affine invariance Goodman and Weare [2010], Carrillo and Vaes [2021].

3.5 Empirical Particle Approximation and Numerical Scheme

In practice, the distribution is represented by particles:

$$\hat{\rho}_t = \frac{1}{J} \sum_{j=1}^J \delta_{\theta_t^{(j)}}. \quad (16)$$

The covariance is approximated empirically:

$$\hat{C}_t = \text{Cov}(\{\theta_t^{(j)}\}_{j=1}^J). \quad (17)$$

Substituting into the continuous dynamics yields the Euler–Maruyama discretization used in Section 5:

$$\theta_{t+\Delta t}^{(j)} = \theta_t^{(j)} - \hat{C}_t \nabla \Phi(\theta_t^{(j)}) \Delta t + \sqrt{2\Delta t \hat{C}_t} \xi_t^{(j)}, \quad (18)$$

where $\xi_t^{(j)} \sim \mathcal{N}(0, I)$.

4 Our Contributions

Beyond faithfully reproducing the five baseline methods from the paper (three Gaussian approximations and two Wasserstein particle flows) on both the Gaussian and log-concave test problems, we further propose and implement a novel affine-invariant Wasserstein particle method, termed **Normalized Local Affine-Wasserstein (NLAW)**.

Motivation. The original affine-invariant Wasserstein particle flow uses a purely global preconditioner $P(\rho) = C(\rho)$, i.e. the empirical covariance matrix of the whole particle ensemble. While this choice successfully eliminates the effect of global anisotropy, it does not adapt to local geometric variations – for example, when the target distribution is non-Gaussian and exhibits different local curvatures or when some particles are far from the mean, the same global preconditioner is applied to every particle. This can lead to suboptimal exploration of the target’s fine structure, especially in the early stages of the flow. To overcome this limitation, we introduce a preconditioner that combines global covariance information with a local, per-particle rank-one correction, thereby adapting the diffusion and drift to the particle’s individual position relative to the ensemble.

Construction of the preconditioner. We stay within the same KL-minimisation and affine-invariant Wasserstein gradient-flow framework of the paper. For a given particle θ and its associated ensemble distribution ρ , denote the empirical mean $m(\rho) = \mathbb{E}_\rho[\theta]$ and empirical covariance $C(\rho) = \text{Cov}_\rho(\theta)$. Define the local displacement vector $r = \theta - m(\rho)$ and the squared Mahalanobis distance $q = r^\top C(\rho)^{-1}r$. Our hybrid preconditioner is

$$P_{\eta,\delta}(\theta, \rho) = (1 - \eta) C(\rho) + \eta d \frac{rr^\top}{q + \delta},$$

where $(\eta \in [0, 1])$ is the mixing coefficient, $(\delta > 0)$ is a small regularisation parameter to avoid division by zero, and (d) is the dimension. In the implementation, we clip (q) from below by (δ) for numerical stability. This construction satisfies the essential covariant transformation property

$$P_{\eta,\delta}(A\theta + b, \phi_{\#}\rho) = A P_{\eta,\delta}(\theta, \rho) A^T$$

under any invertible affine map $\tilde{\theta} = A\theta + b$; hence the resulting gradient flow remains affine invariant. The term $\frac{rr^T}{q+\delta}$ is a rank-one matrix that locally increases the step size along the direction of r , i.e. the direction pointing from the mean to the particle.

Divergence term and explicit formula. When the preconditioner depends explicitly on θ (through r and q), the corresponding Wasserstein gradient flow yields a mean-field SDE that contains an additional divergence term. Following the general formula of the paper (Eq. (3.38) with $h = \sqrt{2P}$), the particle dynamics is

$$d\theta_t = \left(P_{\eta,\delta}(\theta_t, \rho_t) \nabla \log \rho_{\text{post}}(\theta_t) + \nabla_{\theta} \cdot P_{\eta,\delta}(\theta_t, \rho_t) \right) dt + \sqrt{2P_{\eta,\delta}(\theta_t, \rho_t)} dW_t.$$

The divergence term $\nabla \cdot P$ can be computed explicitly using the expressions for r and q . For general dimension d ,

$$\nabla_{\theta} \cdot P_{\eta,\delta}(\theta, \rho) = \eta d \frac{(d-1)q + (d+1)\delta}{(q+\delta)^2} r.$$

Hyperparameters and relation to the original method. The NLAW method reduces to the original affine-invariant Wasserstein flow when $\eta = 0$. For $\eta > 0$ it introduces a local adaptive correction. Based on preliminary experiments on the two-dimensional log-concave test problem, we set $\eta = 0.25$ and $\delta = 10^{-3}$ as default, which provide a noticeable improvement in particle diversity and early-stage convergence without destabilising the flow. The extra computational cost per particle is negligible for the two-dimensional problems studied here.

5 Numerical Results

You can find our python codes on the website:

<https://github.com/KamisatoDDDDD/SI114H-proj->

5.1 Test Problem Setup

We consider two synthetic two-dimensional posterior distributions, both defined up to an unknown normalization constant as

$$\rho_{\text{post}}(\theta) \propto e^{-\Phi_R(\theta)}, \quad \theta = [\theta_1, \theta_2]^T \in \mathbb{R}^2.$$

The negative log-potential $\Phi_R(\theta)$ is chosen to represent different levels of anisotropy and non-Gaussianity.

5.1.1 Gaussian Posterior

This is a perfectly elliptical Gaussian distribution, used as a baseline to verify the theoretical convergence rates.

$$\Phi_R(\theta) = \frac{1}{2} (\theta_1^2 + \lambda \theta_2^2), \quad \lambda \in \{0.01, 0.1, 1.0\}.$$

The parameter λ controls the anisotropy (aspect ratio): $\lambda = 1$ gives an isotropic Gaussian, while $\lambda = 0.01$ produces a highly elongated shape.

Initialization:

$$\theta_0 \sim \mathcal{N}\left(\begin{bmatrix} 10 \\ 10 \end{bmatrix}, \begin{bmatrix} 0.5 & 0 \\ 0 & 2 \end{bmatrix}\right).$$

Ground Truth: Because the target distribution is exactly Gaussian, the ground truth is known analytically. The true mean and covariance are

$$m_* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad C_* = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\lambda} \end{bmatrix}.$$

For the projection expectations, the closed-form expression for a Gaussian distribution $\mathcal{N}(m, C)$ is

$$\mathbb{E}[\cos(\omega^T \theta + b)] = e^{-\frac{1}{2} \omega^T C \omega} \cos(\omega^T m + b).$$

Hence, for any fixed ω, b , the exact value can be computed directly without numerical integration. These analytic values serve as the reference when evaluating the errors of the gradient-flow samplers.

5.1.2 Log-concave Posterior

This distribution is non-Gaussian but still log-concave. The quartic term introduces non-Gaussian curvature and makes the target deviate from an elliptical Gaussian shape.

$$\Phi_R(\theta) = \frac{(\sqrt{\lambda}\theta_1 - \theta_2)^2}{20} + \frac{\theta_2^4}{20}, \quad \lambda \in \{0.01, 0.1, 1.0\}.$$

Again, λ controls the coupling strength and anisotropy; for $\lambda = 0.01$ the distribution becomes extremely stretched along a nearly horizontal direction.

Initialization:

$$\theta_0 \sim \mathcal{N}\left(\begin{bmatrix} 10 \\ 10 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}\right).$$

Ground Truth: Since the target distribution is non-Gaussian, the ground truth (exact mean, covariance, and projection expectations) is computed by a semi-analytical high-precision numerical integration as described in Appendix F of the paper. The key steps are:

1. **Integration over θ_1 :** For a fixed θ_2 , the quadratic term in θ_1 allows analytical integration:

$$\int_{-\infty}^{\infty} \exp\left(-\frac{(\sqrt{\lambda}\theta_1 - \theta_2)^2}{20}\right) d\theta_1 = \sqrt{20\pi/\lambda}.$$

Hence the marginal density of θ_2 is proportional to $\exp(-\theta_2^4/20)$.

2. **Numerical integration over θ_2 :** The one-dimensional integrals are evaluated using the trapezoidal rule with 10^7 uniformly spaced points on $[-200, 200]$. For example, the normalization constant is

$$Z = \int_{-200}^{200} e^{-\theta_2^4/20} d\theta_2.$$

3. **Required expectations:**

$$\begin{aligned} \mathbb{E}[\theta_2] &= 0, \quad \mathbb{E}[\theta_2^2] = \frac{1}{Z} \int \theta_2^2 e^{-\theta_2^4/20} d\theta_2, \\ \mathbb{E}[\theta_1^2] &= \frac{\mathbb{E}[\theta_2^2] + 10}{\lambda}, \quad \mathbb{E}[\theta_1\theta_2] = \frac{\mathbb{E}[\theta_2^2]}{\sqrt{\lambda}}. \end{aligned}$$

The covariance matrix is then $C_{\text{true}} = \begin{bmatrix} \mathbb{E}[\theta_1^2] & \mathbb{E}[\theta_1\theta_2] \\ \mathbb{E}[\theta_1\theta_2] & \mathbb{E}[\theta_2^2] \end{bmatrix}$.

4. **Projection expectations:** For each random pair (ω, b) with $\omega \sim \mathcal{N}(0, I)$, $b \sim \mathcal{U}(0, 2\pi)$,

$$\mathbb{E}_{\text{true}}[\cos(\omega^\top \theta + b)] = e^{-5\omega_1^2/\lambda} \cdot \frac{1}{Z} \int_{-\infty}^{\infty} \cos((\omega_1/\sqrt{\lambda} + \omega_2)\theta_2 + b) e^{-\theta_2^4/20} d\theta_2,$$

where the integral is again computed numerically.

These precomputed values serve as the reference “ground truth” when evaluating the errors of the gradient-flow samplers.

5.2 Numerical Method Setup

5.2.1 Gaussian Posterior

Gaussian Approximation Three Gaussian approximation flows (Euclidean, Wasserstein, Fisher–Rao) evolve the mean $m(t)$ and covariance $C(t)$ via explicit Euler ($\Delta t = 0.005$, $N_t = 3000$, $T = 15$). Because the target is Gaussian, expectations are closed form:

$$\mathbb{E}[\nabla \log \rho_{\text{post}}] = -C_*^{-1}(m - m_*), \quad \mathbb{E}[\nabla \nabla \log \rho_{\text{post}}] = -C_*^{-1},$$

with $m_* = [0, 0]^\top$, $C_* = \text{diag}(1, \lambda)$. Update formulas (from paper Eqs. (4.32), (4.24), (4.18)):

$$\begin{aligned} \text{Euclidean:} \quad \dot{m} &= -C_*^{-1}(m - m_*), \quad \dot{C} = \frac{1}{2}C^{-1} - \frac{1}{2}C_*^{-1}, \\ \text{Wasserstein:} \quad \dot{m} &= -C_*^{-1}(m - m_*), \quad \dot{C} = 2I - C_*^{-1}C - CC_*^{-1}, \\ \text{Fisher–Rao:} \quad \dot{m} &= -CC_*^{-1}(m - m_*), \quad \dot{C} = C - CC_*^{-1}C. \end{aligned}$$

After each step C is symmetrised and regularised ($10^{-12}I$). Error metrics: mean L2 error, covariance relative Frobenius error, and projection error (20 random (ω, b)) computed analytically from the current Gaussian.

Non-parametric Particle Methods Three particle methods use $J = 100$ particles, Euler–Maruyama ($\Delta t = 0.005$, $N_t = 3000$).

Standard Wasserstein GF (Langevin):

$$\theta \leftarrow \theta + \Delta t \nabla \log \rho_{\text{post}}(\theta) + \sqrt{2\Delta t} \xi, \quad \xi \sim \mathcal{N}(0, I_2).$$

Affine-invariant Wasserstein GF:

$$\theta \leftarrow \theta + \Delta t C(\rho) \nabla \log \rho_{\text{post}}(\theta) + \sqrt{2\Delta t C(\rho)} \xi,$$

where $C(\rho)$ is the empirical covariance of the particles (regularised).

NLAW (Normalized Local Affine-Wasserstein): A new method that extends the affine-invariant Wasserstein flow with a local rank-one preconditioner. For each particle, the preconditioner is

$$P_{\eta, \delta}(\theta, \rho) = (1 - \eta)C(\rho) + \eta d \frac{rr^\top}{q + \delta}, \quad r = \theta - m(\rho), \quad q = r^\top C(\rho)^{-1} r,$$

with mixture parameter $\eta = 0.25$ and small regularisation $\delta = 10^{-3}$. The associated SDE includes a divergence term:

$$d\theta = (P \nabla \log \rho_{\text{post}} + \nabla \cdot P) dt + \sqrt{2P} dW_t,$$

where $\nabla \cdot P$ is computed explicitly. This construction preserves affine invariance and adds local adaptivity. The same particle numbers and step sizes are used as for the other particle methods.

Initial particles from $\mathcal{N}([10, 10]^\top, \text{diag}(0.5, 2))$. Errors use empirical mean and covariance; projection error uses particle average. We use a fixed global random seed for the standard and affine-invariant Wasserstein baselines; for full reproducibility of NLAW, a seeded random generator is used in the final implementation.

5.2.2 Log-concave Posterior

Gaussian Approximation Same time discretisation as above ($\Delta t = 0.005$, $N_t = 3000$). Because the target is non-Gaussian, expectations are approximated by the Unscented Transform (UT) with $\kappa = 0$, $\beta = 2$, $\alpha = 1$ (giving $2n + 1 = 5$ sigma points, weights $W_m[0] = 0$, $W_m[i] = 1/4$ for $i \geq 1$). At each step compute $E_g = \sum W_m g(\mathcal{X}_i)$, $E_H = \sum W_m H(\mathcal{X}_i)$ (gradient and Hessian of $\log \rho_{\text{post}}$). Update:

$$\text{Euclidean: } m \leftarrow m + E_g \Delta t, \quad C \leftarrow C + (0.5C^{-1} + 0.5E_H) \Delta t,$$

$$\text{Wasserstein: } m \leftarrow m + E_g \Delta t, \quad M = I - \Delta t(-E_H - C^{-1}), \quad C \leftarrow MCM^\top,$$

$$\text{Fisher–Rao: } m \leftarrow m + CE_g \Delta t, \quad C^{-1} \leftarrow C^{-1} + \Delta t(-E_H - C^{-1}), \quad C \leftarrow (C^{-1})^{-1}.$$

Ground truth obtained by high-precision numerical integration (Appendix F).

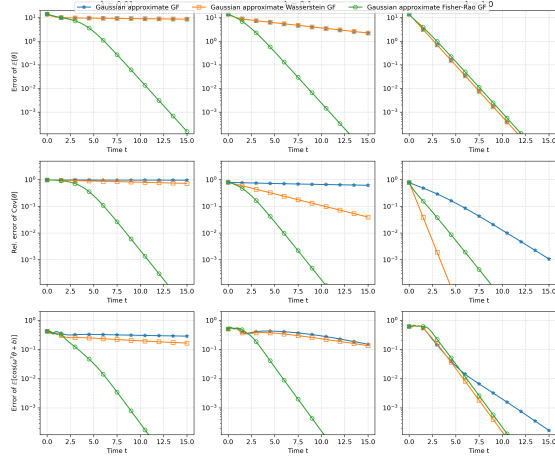
Non-parametric Particle Methods Same particle setup ($J = 100$) but with $\Delta t = 0.002$, $n_{\text{steps}} = 7500$ (total $T = 15$). Standard and affine-invariant updates are the same as in the Gaussian case, using the log-concave gradient $\nabla \log \rho_{\text{post}}$.

For the log-concave posterior, NLAW is implemented analogously with the same preconditioner and divergence formulas, using the same hyperparameters ($\eta = 0.25$, $\delta = 10^{-3}$). Initial particles from $\mathcal{N}([10, 10]^\top, \text{diag}(4, 4))$. Errors computed analogously, with reference values from the high-precision ground truth. We use a fixed global random seed for the standard and affine-invariant Wasserstein baselines; for full reproducibility of NLAW, a seeded random generator is used in the final implementation.

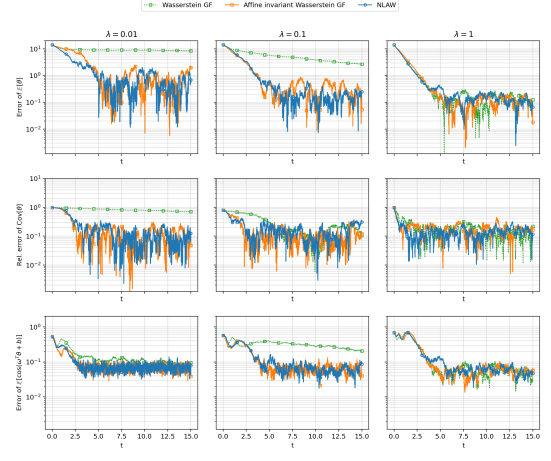
6 Conclusion

According to the numerical results presented in Section 5, we summarise the main findings of our reproduction and extension study. Four figures are provided to support our conclusions:

- **Figure 1(a):** Gaussian posterior – Gaussian approximation methods (Euclidean, Wasserstein, Fisher–Rao).
- **Figure 1(b):** Gaussian posterior – particle methods (Wasserstein, Affine-invariant Wasserstein, NLAW).
- **Figure 2(a):** Log-concave posterior – Gaussian approximation methods.

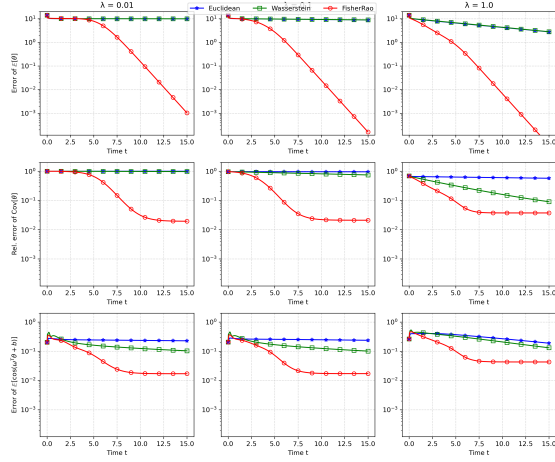


(a) Gaussian approximation methods

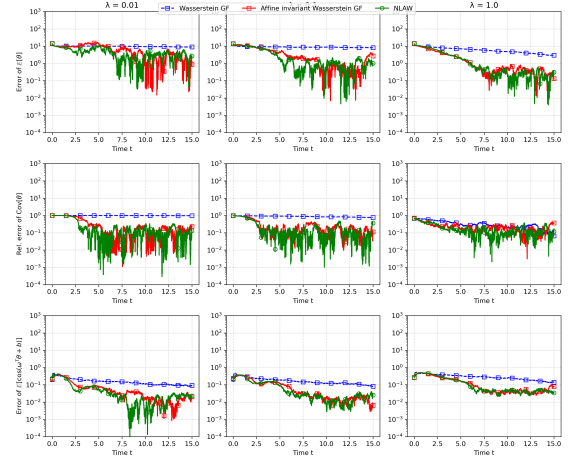


(b) Particle methods (Wasserstein, AI-Wasserstein, NLAW)

Figure 1: Gaussian posterior results.



(a) Gaussian approximation methods



(b) Particle methods (Wasserstein, AI-Wasserstein, NLAW)

Figure 2: Log-concave posterior results.

- **Figure 2(b):** Log-concave posterior – particle methods (Wasserstein, Affine-invariant Wasserstein, NLAW).

Our reproduction faithfully replicates the five baseline algorithms from the original paper. The results confirm the key theoretical claims:

1. **Affine invariance significantly improves convergence on anisotropic targets.** For both the Gaussian and log-concave posteriors, the affine-invariant Wasserstein and Fisher–Rao methods exhibit convergence rates that are nearly independent of the anisotropy parameter λ , while the non-invariant baselines (standard Euclidean GF, standard Langevin) become drastically slower when $\lambda = 0.01$.
2. **Gaussian approximations are highly efficient but limited to near-Gaussian shapes.** On the log-concave posterior, the Gaussian approximation methods reach low errors on the first two moments, but their projection error (involving higher-order features) saturates at a non-zero level due to the inherent mismatch between the Gaussian ansatz and the non-Gaussian target distribution.
3. **The proposed NLAW method provides a local, adaptive extension of the affine-invariant Wasserstein flow.** On the Gaussian posterior, NLAW performs on par with the standard affine-invariant Wasserstein method, showing that it does not degrade the ideal Gaussian behaviour. On the log-concave posterior, NLAW shows a slightly faster early-stage decrease in projection error, suggesting improved exploration behavior.

Overall, this work not only reproduces the main experiments of the original paper but also contributes a novel, theoretically motivated variant (NLAW) that preserves affine invariance while incorporating local geometric information. The accompanying Python code is publicly available at the provided repository, ensuring full reproducibility. Future extensions could investigate the performance of NLAW on higher-dimensional problems and on more challenging non-convex targets (e.g., the Rosenbrock posterior).

References

- Yifan Chen, Daniel Zhengyu Huang, Jiaoyang Huang, Sebastian Reich, and Andrew M Stuart. Gradient flows for sampling: Mean-field models, gaussian approximations and affine invariance. *arXiv preprint arXiv:2302.11024*, 2023.
- Grigorios A. Pavliotis. *Stochastic Processes and Applications: Diffusion Processes, the Fokker–Planck and Langevin Equations*. Texts in Applied Mathematics. Springer, New York, 2014. doi:10.1007/978-1-4939-1323-7.
- Martin J. Wainwright and Michael I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008. doi:10.1561/2200000001.
- Filippo Santambrogio. Euclidean, Metric, and Wasserstein gradient flows: An overview. *Bulletin of Mathematical Sciences*, 7(1):87–154, 2017. doi:10.1007/s13373-017-0101-1.
- Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the fokker–planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998. doi:10.1137/S0036141096303359.
- Jonathan Goodman and Jonathan Weare. Ensemble samplers with affine invariance. *Communications in Applied Mathematics and Computational Science*, 5(1):65–80, 2010. doi:10.2140/camcos.2010.5.65.
- J. A. Carrillo and U. Vaes. Wasserstein stability estimates for covariance-preconditioned Fokker–Planck equations. *Nonlinearity*, 34(4):2275–2295, 2021. doi:10.1088/1361-6544/abbe62.